

iknbite.com

AI Voice Studio

Full Project Documentation

Development Journey & Technical Reference

Date: July 29, 2026

Version: 2.0

Repository: github.com/gsops00/iknbite.github.io

Live Site: <https://iknbite.com>

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1. Project Overview

What is iknbite?

iknbite.com is a free, open-source AI Voice Studio that runs entirely in the browser. It allows users to generate natural-sounding speech from text using multiple TTS (Text-to-Speech) backends. The platform supports 76 voices across 28 languages with features like emotion control, SSML editing, multi-speaker conversations, AI script writing, and image generation.

Key Statistics

- 76 voices across 28 languages
- 8 emotion modes (Happy, Sad, Angry, Calm, Excited, etc.)
- 11 voice presets for different use cases (YouTube, TikTok, Podcast, etc.)
- 83 avatar images (real character portraits)
- AI Chatbot for script writing with 6 template styles
- Image Studio with free text-to-image generation
- Zero server cost - all processing is client-side or uses free APIs

Target Audience

- Content creators (YouTube, TikTok, Podcasts)
- Voice-over artists and narrators
- Game developers needing quick voice lines
- Language learners
- Accessibility users who need text read aloud

2. Architecture

System Design

iknbite follows a static-site architecture with client-side JavaScript. All rendering and TTS logic runs in the browser. The site is hosted on GitHub Pages for zero-cost hosting. TTS processing can use local browser APIs (Web Speech) or external services (ElevenLabs API, Edge TTS via Cloudflare Worker, EasyVoice/Kokoro API).

Data Flow

User selects a voice and enters text

1. TTS Engine checks available backends in order:
 - a) ElevenLabs API (highest quality, requires API key)
 - b) EasyVoice/Kokoro API (requires API key)
 - c) Edge TTS via Cloudflare Worker (free, recommended)
 - d) Browser Web Speech API (fallback, limited quality)
2. Audio is generated as a blob
3. User can play, download, or share the audio

Key Decisions

- Static HTML + JS (no build step needed for main site)
- GitHub Pages for hosting (free, unlimited bandwidth)
- Single-page application (SPA) with hash-based navigation
- All state in JavaScript object + localStorage
- Modular JS files for maintainability
- CSS custom properties for theming (dark/light mode)

3. Technology Stack

Frontend

- Vanilla JavaScript (ES6+) - no framework overhead
- CSS3 with Custom Properties - theming system
- HTML5 - semantic structure
- Google Fonts: Inter, Manrope, Syne, Noto Sans Arabic, Tajawal
- GSAP (GreenSock) - scroll animations

TTS Backends

- ElevenLabs API - premium neural voices (requires API key)
- Edge TTS (Microsoft) - 300+ neural voices, 80+ languages
- EasyVoice/Kokoro API - open-source TTS
- Web Speech API - browser built-in (fallback)

APIs Used

- ElevenLabs TTS API - text-to-speech generation
- Pollinations.ai - free text-to-image generation
- MyMemory Translation API - free text translation
- Hugging Face Inference API - AI chatbot (free tier)

Hosting

- GitHub Pages - primary hosting (free)
- Cloudflare Workers - Edge TTS proxy (free tier)

Dev Tools

- Git - version control
- Composio CLI - GitHub integration tools
- Wrangler CLI - Cloudflare Workers deployment

Audio

- Web Audio API - audio playback and processing
- Blob/URL API - audio download management

4. Complete File Structure

```
iknbite/
|-- index.html           # Main HTML entry point
|-- 404.html             # Custom 404 page
|-- CNAME                # Custom domain (iknbite.com)
|-- start.sh             # Local dev server script
|-- edge-tts-worker.js   # Cloudflare Worker for Edge TTS
|
|-- css/
|   +-- style.css        # Complete styling (1009 lines)
|
|-- js/
|   |-- avatars.js       # Avatar system (17 lines)
|   |-- features.js      # Emotions, SSML, Translation (336 lines)
|   |-- voices.js        # 76 voice definitions (172 lines)
|   |-- chatbot.js       # AI script assistant (243 lines)
|   |-- tts.js           # TTS engine core (716 lines)
|   |-- ui.js            # UI rendering (1966 lines)
|   +-- app.js           # App entry & shell (205 lines)
|
|-- img/avatars/         # 83 character portrait images
|   |-- ava.jpg          # Female American voice
|   |-- andrew.jpg       # Male American voice
|   +-- ...              # (83 images total)
|
|-- tts-engine/          # Voice training pipeline
|   |-- config.yaml      # Training configuration
|   |-- requirements.txt  # Python dependencies
|   +-- scripts/         # Training scripts
|
|-- v2/                  # React/Next.js version (WIP)
|   |-- package.json     # Dependencies
|   |-- tsconfig.json    # TypeScript config
|   +-- src/             # React source files
|
+-- .git/                # Git repository
```

5. Features

Voice Studio

The core feature. Select a voice, type text, adjust speed/pitch/emotion, and generate speech. Supports SSML and multi-speaker.

Voice Library

Browse 76 voices with search, filter by language/gender/category, sort by popularity/quality/name.

Image Studio

Free text-to-image generation using Pollinations.ai. No login required.

AI Chatbot

Script writing assistant with 6 templates: Narration, Story, Podcast, Educational, Commercial, Custom.

Emotion System

8 emotions: Happy, Sad, Angry, Calm, Excited, Serious, Fear, Romantic. Each adjusts pitch and speed.

SSML Builder

Speech Synthesis Markup Language - pauses, emphasis, pronunciation control.

Smart Recommender

AI analyzes text and recommends the best voice, emotion, and speed settings.

Translation

Translate text between 28 languages using MyMemory API before generating speech.

Multi-Speaker

Format as "SpeakerName: dialogue" for multi-voice conversations.

Presets

11 presets: YouTube, TikTok, Podcast, Audiobook, Education, etc.

Download

Download audio as MP3. Multiple strategies for cross-browser compatibility.

Dark/Light Mode

Full theme support with smooth transitions.

Voice Training

AnyClaw engine - define archetypes, emotion profiles, and datasets.

6. Voice Engine Details

TTS Engine Flow

TTSEngine.speak(text, voiceDef, options):

1. ElevenLabs (highest quality) - POST `api.elevenlabs.io/v1/text-to-speech/{voiceId}`
2. EasyVoice/Kokoro - POST `{url}/v1/audio/speech`
3. Edge TTS (Microsoft Neural) - POST `{apiUrl}/tts` (300+ voices, 80+ languages)
4. Web Speech API (fallback) - Browser built-in

Voice Mappings

Each iknbite voice ID maps to different backend IDs:

- `EDGE_VOICE_MAP`: Microsoft Edge TTS neural voice IDs (en-US-AvaNeural, ar-SA-HamedNeural, etc.)
- `EL_VOICE_MAP`: ElevenLabs pre-made voice IDs (21m00Tcm4TlVDq8ikWAM for Rachel, etc.)
- `EASYVOICE_MAP`: Kokoro TTS voice IDs (af_aoede, af_bella, etc.)

Download Strategies (Mobile)

4 download strategies in order:

- S1: File System Access API (`showSaveFilePicker`) - Chrome 86+ Android
- S2: Blob URL + `<a click>` - Desktop browsers
- S3: Chunked Base64 Data URL - 8KB chunks to avoid crash
- S4: Floating download button - When popup is blocked

7. Development History (Git Commits)

2026-07-29	fix: improve text contrast, fix mobile download, fix avatar fallbacks
2026-07-29	feat: improve hero section - Image Studio button, fix stats
2026-07-29	fix: add missing commas between UI methods - fixes white screen
2026-07-29	fix: white screen - fix escaped backticks/dollar signs in templates
2026-07-29	feat: add Image Studio with Pollinations.ai integration
2026-07-29	fix: improve mobile download with multi-strategy fallback
2026-07-29	feat: embed ElevenLabs API key for instant voice generation
2026-07-28	fix: apply previous improvements (chatbot, voices, UI, accessibility)
2026-07-27	fix: darker text for better readability
2026-07-27	fix: missing commas in voices array + broken quotes in onclick handlers
2026-07-27	feat: premium voice library redesign - search, filters, sorting
2026-07-26	fix: avatar images not loading + improve voice card name visibility
2026-07-26	fix: push all avatar images + remove HeyGen + fix waveform overflow
2026-07-25	feat: glassmorphism redesign + GSAP animations + HeyGen integration
2026-07-25	feat: add ElevenLabs TTS backend with voice mapping
2026-07-25	feat: rebuild Features page - Emotion Tester, SSML Builder, Translation
2026-07-24	feat: add AI features module + enhanced Studio with emotions/presets
2026-07-24	feat: Add Training page
2026-07-22	feat: Add VCTK Corpus dataset integration
2026-07-21	feat: Add Mozilla Common Voice dataset integration

8. Bugs Fixed

White Screen (Critical)

Site showed blank white page. Cause: Missing commas between UI object methods and unescaped template literals causing JS parse errors. Fix: Added commas between 4 methods and escaped special characters in template literals.

Download Not Working on Mobile

Download button did nothing on Chrome Android. Cause: blob URL + <a download> not supported on Chrome Android. Also btoa() on large audio crashed due to JS argument limits. Fix: 4 strategies with File System Access API as primary, chunked base64 (8KB chunks), floating button as fallback.

Text Contrast Too Low

Text too light in light mode. "al-kitaba ghayr wadiha" (writing not clear). Cause: --surface-400 at 60% opacity. Fix: Increased to 72%, 78%, 85% for surface-400/500/600.

Avatars Without Names

Voice cards showed images without names. Cause: generateAvatar() used voice.id without null check. Fix: Added defensive checks.

9. Sources & Libraries

Open Source Libraries

GSAP (GreenSock Animation)

cdnjs.cloudflare.com/ajax/libs/gsap/3.12.5/ - Animations

Google Fonts

fonts.googleapis.com/ - Inter, Manrope, Syne, Arabic fonts

Composio CLI

composio.dev/ - GitHub integration

APIs Used

ElevenLabs TTS

api.elevenlabs.io/v1/text-to-speech/{voice} - Premium TTS (key required)

Pollinations.ai

pollinations.ai/prompt/{prompt} - Free image generation

MyMemory Translation

api.mymemory.translated.net/get - Free translation

Hugging Face

api-inference.huggingface.co/models/ - AI chatbot (free tier)

Edge TTS

Via Cloudflare Worker - 300+ free neural voices

Voice Datasets (Training)

LibriTTS	English	585h	CC0
VCTK	English	44h	CC0
LJSpeech	English	24h	CC0
Common Voice 17	Multilingual	4000h+	CC0
Arabic Speech	Arabic	3.7h	CC BY 4.0
MASC	Arabic	10h	CC BY
ESD	English	10h	Academic

RAVDESS	English	8h	CC BY
CREMA-D	English	6h	CC BY

10. How to Deploy

GitHub Pages

```
# Commit and push
cd /root/iknbite
git add -A
git commit -m "description"
git push origin main
# Site live at https://iknbite.com (1-2 min)
```

Deploy Edge TTS (Cloudflare Worker)

For real male/female voice differentiation:

```
cd /root/cloudflare-edge-tts
npx wrangler deploy
# Then paste URL in iknbite Settings page
```

Run Locally

```
# Python server
cd /root/iknbite
python3 -m http.server 8080
# Open: http://localhost:8080

# Or with start script
./start.sh 8080
```

Checklist After Deploy

- Verify all pages load (Home, Studio, Voices, Images, Settings, Chat)
- Test voice selection and generation
- Test download on mobile Chrome
- Check dark/light mode toggle
- Verify avatar images for all 76 voices
- Test text readability in light mode

11. Future Roadmap

Batch Download

Generate multiple voice variations as ZIP

Voice Cloning (Beta)

Upload 10-30s sample for custom voice

Script Assistant

AI helper for TTS-optimized narration scripts

API Dashboard

API key generation for future monetization

Enhanced AI Chatbot

Better generation with Fable 5 prompts

i18n

Full Arabic localization and RTL improvements

Analytics

Voice usage tracking and popularity metrics

More Avatars

Anime, 3D, realistic avatar styles

Performance

Service worker for offline, lazy loading

Technical Debt

- v2/ React version incomplete - needs finishing or removal
- 7 extra avatar images without matching voice definitions
- ElevenLabs API key hardcoded (security concern - use env vars)
- _shareAudio indentation inconsistent (cosmetic)
- No error tracking for production monitoring
- No automated tests